



Quartet Filtering for Likelihood Mapping in IQ-Tree

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Project Supervisor: Heiko Schmidt

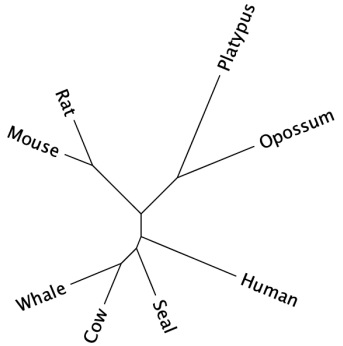
Universität Wien

13.11.2026

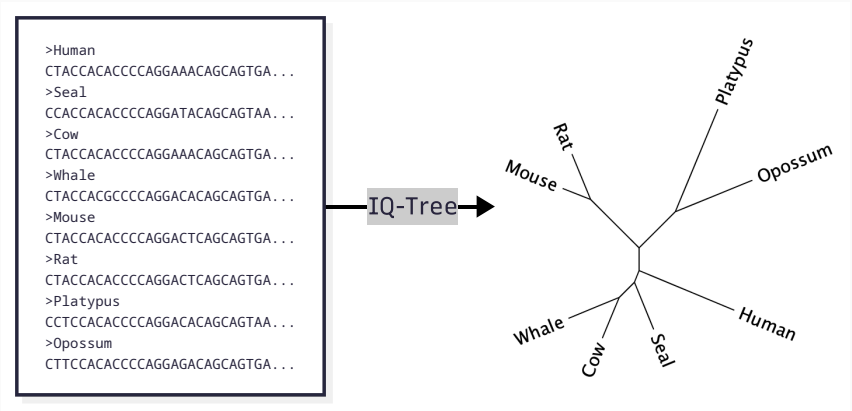
Background - Phylogenetic Inference

```
>Human
CTACCACACCCCAGGAAACAGCAGTGA...
>Seal
CCACCACACCCCAGGATACAGCAGTAA...
>Cow
CTACCACACCCCAGGAAACAGCAGTGA...
>Whale
CTACCACGCCCCAGGACACAGCAGTGA...
>Mouse
CTACCACACCCCAGGACTCAGCAGTGA...
>Rat
CTACCACACCCCAGGACTCAGCAGTGA...
>Platypus
CCTCCACACCCCAGGACACAGCAGTAA...
>Opossum
CTTCCACACCCCAGGAGACAGCAGTGA...
```

IQ-Tree →



Background - Phylogenetic Inference



- resulting tree that is best explained by data, i.e. maximum likelihood estimation

  $\mathcal{L}(\text{tree, model of evolution}) = \mathbb{P}(\text{aligned sequences} \mid \text{tree, model of evolution})$

- Phylogenetic information present in alignment?

Quartets and Likelihood Mapping

- Assess whether the alignment is suitable for a phylogenetic reconstruction.

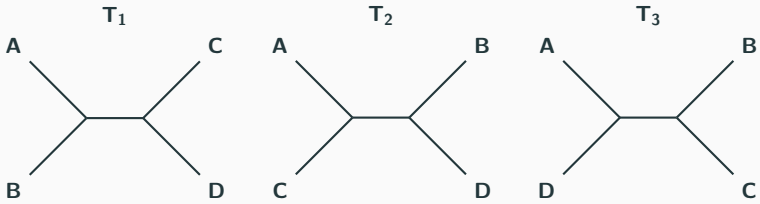
Quartets and Likelihood Mapping

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- Can a tree be reliably resolved, even if its (quartet) subtrees cannot?

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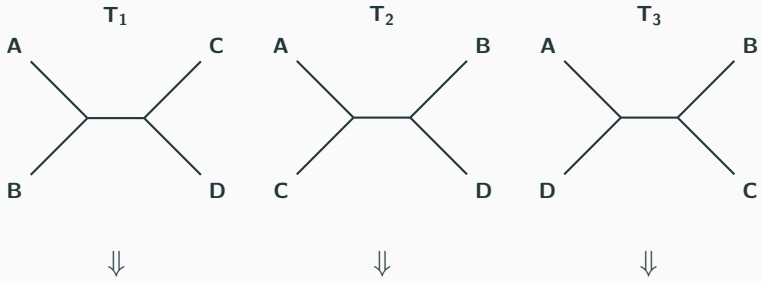
For a quartet $\{A, B, C, D\}$ of leaves, there are only 3 possible tree topologies (up to isomorphism):



Quartets and Likelihood Mapping

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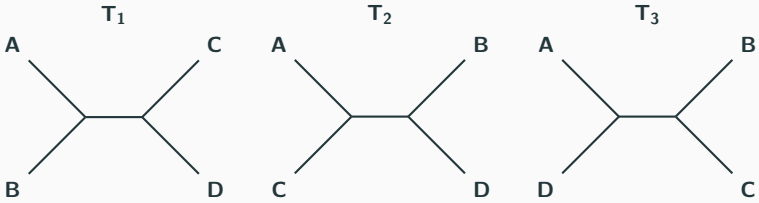


resulting likelihoods L_1 L_2 L_3

Quartets and Likelihood Mapping

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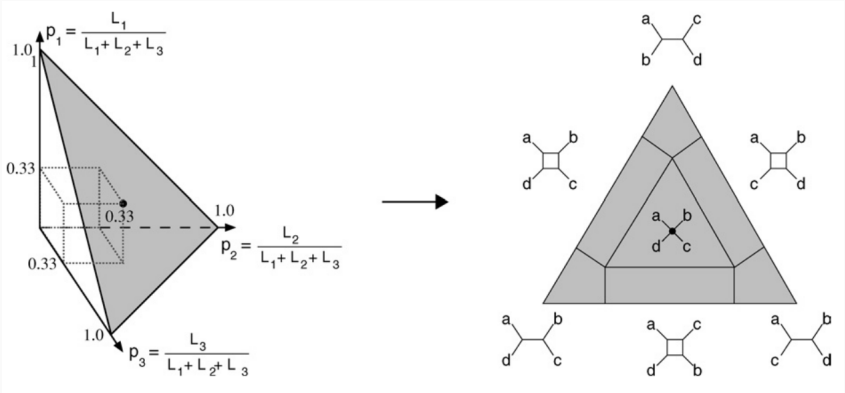


resulting likelihoods	L_1	L_2	L_3
posteriors	$p_1 = \frac{L_1}{L_1+L_2+L_3}$	$p_2 = \frac{L_2}{L_1+L_2+L_3}$	$p_3 = \frac{L_3}{L_1+L_2+L_3}$

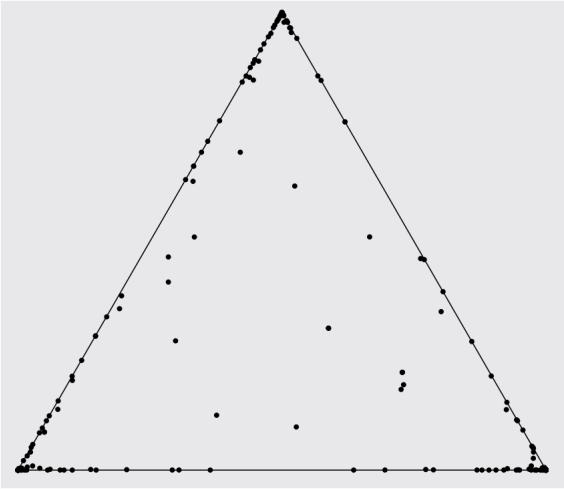
Quartets and Likelihood Mapping Continued

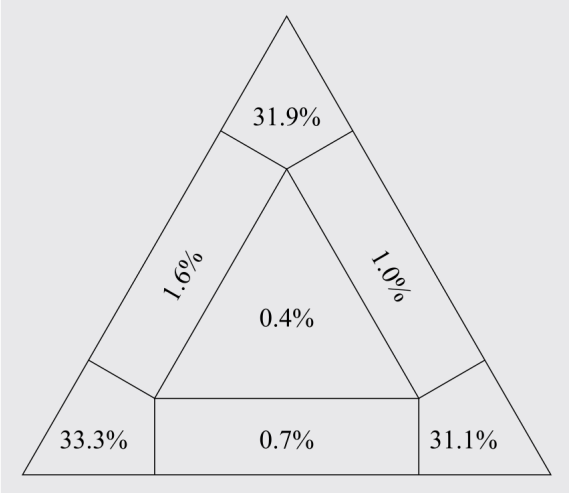
$$p_i = \frac{L_i}{L_1+L_2+L_3}$$

$$p_i \geq 0 \text{ and } p_1+p_2+p_3 = 1 \implies (p_1, p_2, p_3) \in \Delta^2 = \left\{ (x_0, x_1, x_2) \in \mathbb{R}^3 \mid x_i \geq 0, \sum_{i=0}^{2} x_i = 1 \right\}$$

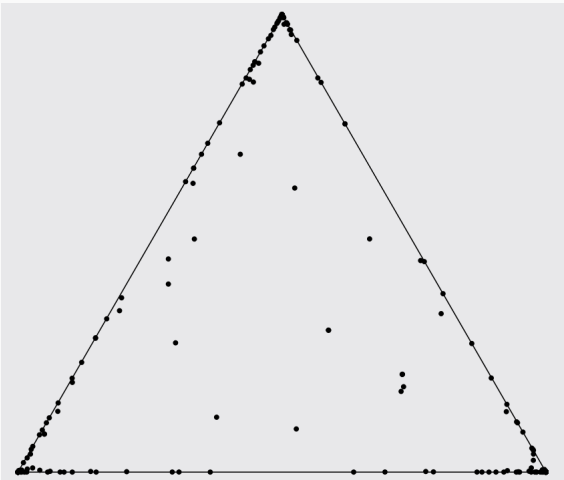


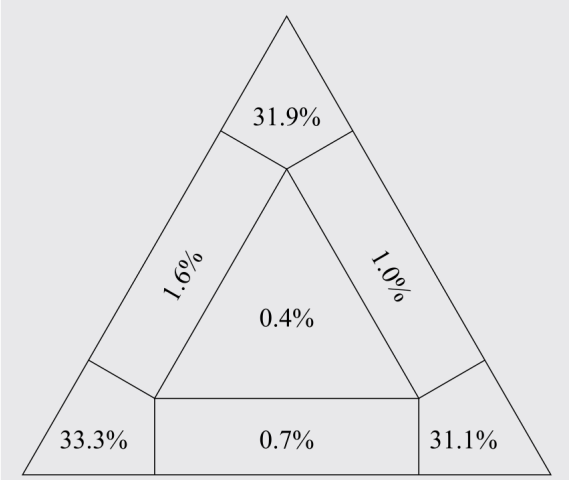
Likelihood Mapping Diagrams





Likelihood Mapping Diagrams



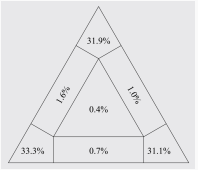


Heuristic: more than $20 - 30\%$ outside of the corners $\Longrightarrow$ data not reliable for phylogenetic inference.

Likelihood Mapping Summary

Quartet resolution per sequence (phylogenetic information):

	#quartets	resolved	partly	unresolved	name
1	276	271 (98.19)	5 (1.81)	0 (0.00)	HIV1groupO
2	291	283 (97.25)	8 (2.75)	0 (0.00)	SIVcpzUS
3	289	272 (94.12)	13 (4.50)	4 (1.38)	SIVcpzGAB
4	295	274 (92.88)	17 (5.76)	4 (1.36)	aU09127
5	286	273 (95.45)	9 (3.15)	4 (1.40)	gU27426
6	267	255 (95.51)	12 (4.49)	0 (0.00)	gU27445
7	263	256 (97.34)	7 (2.66)	0 (0.00)	cAF067158
8	289	269 (93.08)	20 (6.92)	0 (0.00)	cU09126
9	297	288 (96.97)	9 (3.03)	0 (0.00)	dU27399
10	282	271 (96.10)	11 (3.90)	0 (0.00)	dU43386
11	288	283 (98.26)	3 (1.04)	2 (0.69)	bL02317
12	292	283 (96.92)	9 (3.08)	0 (0.00)	bAF025763
13	301	297 (98.67)	2 (0.66)	2 (0.66)	bU08443
14	284	277 (97.54)	7 (2.46)	0 (0.00)	bAF042106
	1000	963 (96.30)	33 (3.30)	4 (0.40)	



Caveats of Likelihood Mapping

1. **Uninformative Quartets**
2. **Quartet Permutations**
3. **Non Overlapping Sequences**

1. Uninformative Quartets

Not all quartets might be equally informative.

Example (Identical Sequences)

```
>Seq_1
CTACCACACCCCAGGAAACAGCAGTGA...
>Seq_2
CTACCACACCCCAGGAAACAGCAGTGA...
>Seq_3
CTACCACACCCCAGGAAACAGCAGTGA...
>Seq_4
CCACCACACCCCAGGATACAGCAGTAA...
```

$$L_1 = L_2 = L_3$$
$$\Longrightarrow$$
$$p_i = \frac{1}{3}$$

- 3 identical sequences cause a quartet to be unresolvable
- by default, 3rd and subsequent identical sequences will be ignored
- however, these occurrences are also ignored in the likelihood mapping table

Improvement

Include all occurrences in likelihood mapping table.

Other Uninformative Quartets

Example

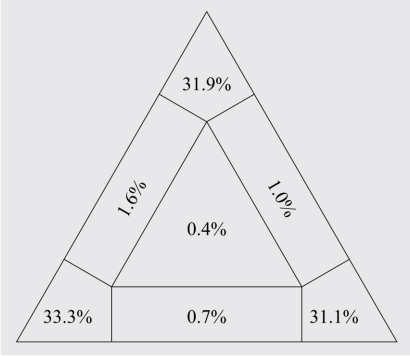
```
seq1: ATACCACACC CCAGGAAACA GCAGTGA
seq2: CTACCACACT CCAGGAAACA GCAGTGA
seq3: CTACCACACC CCAGGAGACA GCAGTGA
seq4: CTACCACACC CCAGGAAACA GCAGGGA
      *        *       *        *
```

- sequences identical up to “singleton” sites

2. Quartet Permutations

`iqtree3 -lmap 1000`

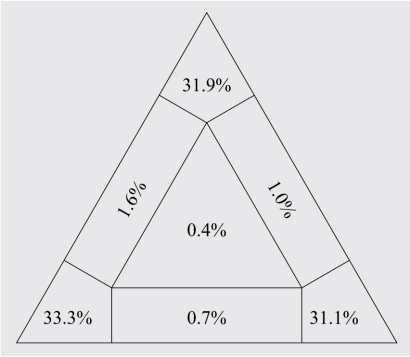
- specifies number of quartets to be drawn
- quartets are drawn uniformly at random



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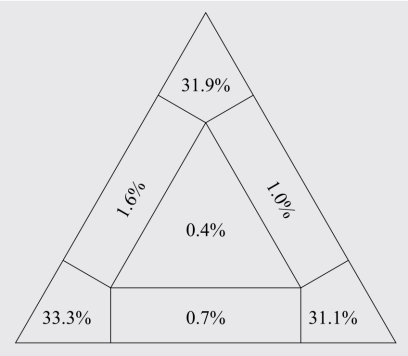
`iqtree3 -lmap ALL`

- all possible $\binom{n}{4}$ quartets are drawn
- in this case $n = 14 \implies \binom{14}{4} = 1001$

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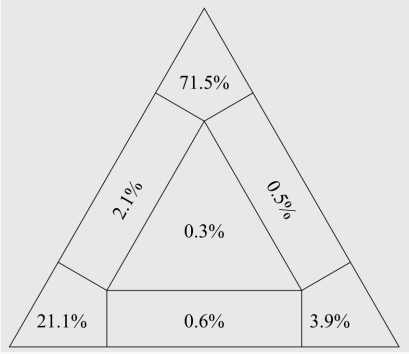
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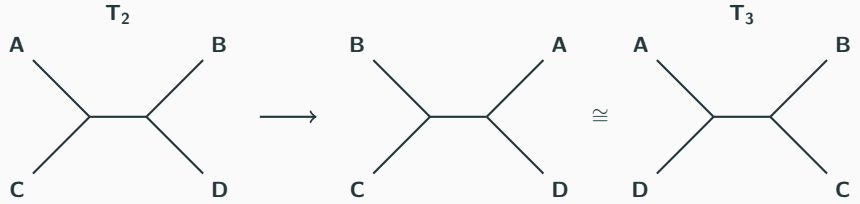


Quartet Permutations - Explanation

- with `-lmap ALL`, sequences in a quartet are drawn according to their order in the alignment
- changing their order changes the tree topology and thus their likelihoods
- mathematically: there is a group homomorphism $S_4 \to \mathrm{Sym}(\{T_1, T_2, T_3\}) \cong S_3$

Example

Swap A and B:

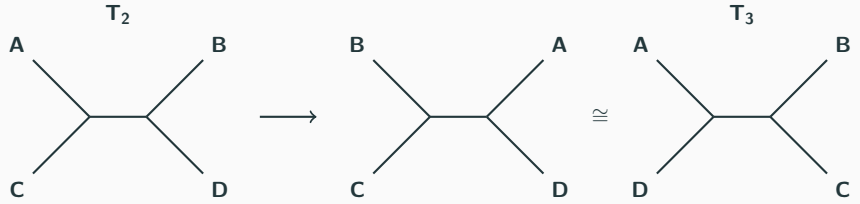


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Example

Swap A and B:



Possible Fix

Permute quartets randomly.

3. Non Overlapping Sequences

```
Seq_1:  TTGTATGTTT GATGCCTCTA TTGTATGTTT GAT
Seq_2:  TTGCATGATT AATGGATCTG TTGCATGATT AAT
Seq_3:  ---------- -------CTA TTGTATGTTT GAT
Seq_4:  TTGCATGATT AATGG----- ---------- ---
Seq_5:  TTGCATGATT AATGGCTCTA TTGTATGTTT GAT
```

$\Longrightarrow$ non-overlapping sequences might cause wrongly resolved or unresolved quartets

Pairwise Completeness Score

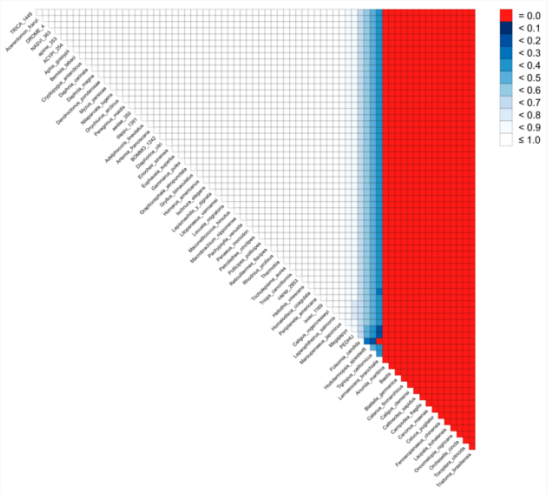
Definition

$$C_{ij} = \frac{\left|\left\{k \mid S_i^k \text{ and } S_j^k \text{ both unambiguous}\right\}\right|}{n}$$

Possible Implementation

Filter by minimum completeness score

- required memory: $\Theta(m^2)$



Implementation Details

- C++ (`-std=c++17`)
- $>$ 80000 lines of code
- CMake Build System
- Dependencies:
  - Eigen3
  - Boost
  - MPI (Message Passing Interface)
- Git repository hosted on GitHub
- Documentation with Doxygen
- Test scripts included
- CI script running on GitHub Actions

"There are only two kinds of languages: the ones people complain about and the ones nobody uses" (Bjarne Stroustrup)

```
template<class T>
concept isBar = requires(T obj, const Span& sp, Bar&& bs) {
    obj.bar(sp, bs);
};
```

MY CONCEPTS LOOK LIKE THIS

```
template<class T>
    requires isBar<T>
void foo(T obj, const Span& sp, Bar&& bs) {
    obj.bar(sp, bs);
}
```

SO HER CODE CAN LOOK LIKE THIS



Thank you!

References

[1] Heiko A. Schmidt and Arndt von Haeseler, 2009, *The Phylogenetic Handbook: A Practical Approach to Phylogenetic Analysis and Hypothesis Testing*, DOI: 10.1017/CBO9780511819049.008.

[2] Korbinian Strimmer and Arndt von Haeseler, 1997, "Likelihood-mapping: A simple method to visualize phylogenetic content of a sequence alignment", *Proceedings of the National Academy of Sciences*, DOI: 10.1073/pnas.94.13.6815.

[3] Thomas K F Wong et al., 2026, "IQ-TREE 3: phylogenomic inference software using complex evolutionary models", *Mol Biol Evol*, DOI: 10.1093/molbev/msag117.